10 CFR 50.90

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U. S. Nuclear Regulatory Commission
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Subject: **Palo Verde Nuclear Generating Station (PVNGS) Units 1, 2, and 3**
Docket Nos. STN 50-528, 50-529, and 50-530
Renewed Operating License Nos. NPF-41, NPF-51, and NPF-74
License Amendment Request to Modify Required Actions for Reactor Coolant System (RCS) Leak Detection Instrumentation

Pursuant to 10 CFR 50.90, Arizona Public Service Company (APS) is submitting a request for an amendment to the Technical Specifications (TS) for Palo Verde Nuclear Generating Station (PVNGS) Units 1, 2, and 3. The proposed change revises TS Section 3.4.16, *RCS Leakage Detection Instrumentation*, to modify the required actions such that a limited time period is provided to restore RCS leak detection instrumentation, in lieu of plant shutdown, so long as alternative RCS leak detection actions are implemented.

A license amendment request (LAR) pre-submittal meeting was held on February 10, 2026 [NRC Agencywide Documents Access and Management System (ADAMS) Accession Number ML26033A039]. The enclosure to this letter provides a description of the proposed changes, a technical evaluation, a regulatory evaluation including a no significant hazards consideration, and an environmental assessment. The enclosure is supported by three attachments.

Attachment 1 of the enclosure provides a marked-up existing TS page. Attachment 2 of the enclosure provides a revised (clean) TS page. A mark-up of the affected TS Bases pages in support of the TS change is provided for information in Attachment 3. Approval of the proposed amendment is requested by December 18, 2026. Once approved, the amendment is planned to be implemented within 90 days.

In accordance with the PVNGS Quality Assurance Program, the Plant Review Board has reviewed and approved this LAR. By copy of this letter, the LAR is being forwarded to the Arizona Department of Health Services - Bureau of Radiation Control for information.

No new commitments are being made to the NRC by this letter.

Should you need further information regarding this LAR, please contact Michael D. Dilorenzo, Department Leader Nuclear Regulatory Affairs - Licensing, at (623) 393-3495.

I declare under penalty of perjury that the foregoing is true and correct to the best of my knowledge.

Executed on March 4, 2026
(Date)

Sincerely,

Spina, Jennifer (Z08962)

Digitally signed by Spina, Jennifer (Z08962)
Date: 2026.03.04 17:14:15 -07'00'

JLS/MDD/cmr

Enclosure: Description and Assessment of Proposed License Amendment
Attachment 1: Technical Specification Pages Mark-ups
Attachment 2: Clean Technical Specification Pages
Attachment 3: Technical Specification Bases Page Mark-up (provided for information only)

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ENCLOSURE

Description and Assessment of Proposed License Amendment

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1. SUMMARY DESCRIPTION

This evaluation supports a request to amend Renewed Facility Operating License Nos. NPF-41, NPF-51, and NPF-74, for Palo Verde Nuclear Generating Station (PVNGS) Units 1, 2, and 3, respectively.

In accordance with the provisions of Section 50.90 of Title 10 of the Code of Federal Regulations (10 CFR), Arizona Public Service Company (APS) is submitting a License Amendment Request (LAR) to revise the Technical Specifications (TS) for PVNGS Units 1, 2, and 3. The LAR proposes revisions to TS Section 3.4.16, *RCS Leakage Detection Instrumentation*, required actions, such that a limited time period is provided to restore reactor coolant system (RCS) leak detection instrumentation, in lieu of plant shutdown, so long as alternative RCS leak detection actions are implemented.

This change is planned to be implemented within 90 days of license amendment (LA) approval.

2. Description of Proposed Technical Specification Changes

2.1 TS 3.4.16 – RCS Leakage Detection Instrumentation

Technical Specification 3.4.16 establishes the limiting condition for operation (LCO) for instrumentation credited for detecting leakage from the RCS during power operations. The RCS leakage detection instrumentation is described in the Updated Final Safety Analysis Report (UFSAR) Section 5.2.5.

Multiple instrument locations are utilized, if needed, to help identify the location of leakage sources. The safety significance of RCS leakage varies widely depending on its source, rate, and duration. Therefore, detecting and monitoring RCS leakage into the containment area is necessary. Quickly separating the identified leakage from the unidentified leakage provides quantitative information to the operators, allowing them to take corrective action should leakage occur detrimental to the safety of the facility and the public. RCS leakage detection instrumentation satisfies Criterion 1 of 10 CFR 50.36 (c)(2)(ii).

2.2 Proposed Change to Required Actions and Completion Times

The current PVNGS LCO 3.4.16 Required Action Condition D, for a loss of all TS credited RCS leak detection instrumentation, is to enter LCO 3.0.3 immediately, which directs that Action shall be initiated within 1 hour to place the unit, as applicable, in:

a. MODE 3 within 7 hours;
b. MODE 5 within 37 hours.

This LAR proposes, that in lieu of immediate plant shutdown, that up to 72 hours be available to restore either the radiation monitoring function or the containment sump leak detection function to operable status, provided alternate RCS leak detection actions are implemented. Specifically, existing Condition C would be redesignated as D and a new Condition C added that contains Required Actions:

C.1 Analyze grab samples of the containment atmosphere.

OR

C.2 Perform SR 3.4.14.1.

AND

C.3 Restore at least one RCS leakage detection monitor to OPERABLE status.

The Completion Times for Required Actions C.1 and C.2 would be once per 12 hours, and the Completion Time for Required Action C.3 would be 72 hours.

A mark-up of the affected TS pages is provided in Attachment 1 to this enclosure. Clean TS pages are provided in Attachment 2 to this enclosure.

A mark-up of the affected TS Bases in support of the TS change is provided for information in Attachment 3.

2.3 Reason for the Proposed Change

The low-risk nature of the loss of the primary RCS leak detection instrumentation, while retaining alternate RCS leak detection actions for a limited time, supports not requiring a rapid plant shutdown, as currently required by TS LCO 3.4.16.

The Nuclear Regulatory Commission (NRC) has encouraged licensees to use risk information where practical to reduce unnecessary conservatism associated with current regulatory requirements. An immediate plant shutdown as required by LCO 3.0.3 could result in an unnecessary transient without a corresponding health and safety benefit. While reactor operators are trained to perform a rapid plant shutdown, moving from full power to cold shutdown is a major plant evolution that exercises an array of plant equipment and procedures. Most TS required equipment is in standby, and its inoperability does not threaten stable plant operation. Therefore, maneuvering the plant through a rapid shutdown transient may be an unwarranted risk if the condition can be resolved in a reasonable period, or failing restoration, for preparation of an orderly shutdown or to request appropriate regulatory relief.

Historical information has shown that about half of the plant shutdowns initiated under LCO 3.0.3 could be avoided if licensees had additional time to resolve the condition or obtain relief from the NRC. A plant shutdown under LCO 3.0.3 is also time-critical in that a major plant evolution must be initiated within one hour. Initiation of a plant shutdown includes many activities, such as stopping maintenance and testing and restoring systems. Experience indicates that time-critical actions are more error prone, and providing additional time to initiate a plant shutdown could relieve time pressure and reduce the risk of human error.

The NRC Notice of Enforcement Discretion guidance (NRC Enforcement Manual, Appendix F, *Notices of Enforcement Discretion*) recognizes that a plant shutdown is not always the safest course of action:

> "The NRC has historically recognized that the two safest modes for operating a nuclear power plant are either Mode 5 (shut down) or Mode 1 (operating at power). Transitions between these two modes may introduce situations or configurations that involve an increase in risk. The NRC expects its licensees to comply with all applicable requirements (i.e., regulations, license conditions, etc.) However, circumstances may arise at an operating [Nuclear Power Plant] NPP where compliance with a TS LCO or a license condition would result in an unnecessary transient without a corresponding health and safety benefit…"

The goal of the proposed change is to avoid an unwarranted plant transient. It is unlikely that a licensee could request and obtain relief from the NRC, such as Enforcement Discretion, or a license amendment, within the existing one-hour delay provided by LCO 3.0.3. However, under the proposed change pursuing such relief during the proposed 72-hour period is possible, in parallel with efforts to correct the condition and to prepare for an orderly shutdown.

Finally, the current requirement to enter LCO 3.0.3 on loss of all RCS leak detection instrumentation provides a disincentive to perform maintenance on the RCS leak detection instrumentation equipment at power, as a failure of related instrumentation during maintenance would require immediate plant shutdown. The proposed change would reduce this disincentive and could ensure better reliability of the instrumentation through timely maintenance.

2.4 Background of the PVNGS RCS Leak Detection System Design and Licensing Basis

While the proposed license amendment does not propose to change the current PVNGS LCO 3.4.16 or Applicability requirements, the following information is provided to give the NRC staff reviewers historical and operational context for the proposed change to the Required Actions and Completion Times for loss of RCS leak detection instrumentation.

When PVNGS was originally licensed, the RCS leak detection system LCO requirements included the noble gas (gaseous) and particulate radiation monitoring and sump instruments; however, either the gaseous or the particulate channel of the radiation monitor (RU-1) could satisfy the original radiation monitoring function for the LCO.

With improvements in fuel performance, such that RCS gaseous concentrations were reduced during power operations, the NRC, industry, and APS recognized the need to not rely solely upon the gaseous channel of RU-1 to satisfy the radiation monitoring RCS leak detection function for the RCS leak detection system LCO.

In LA 117, in 1998, when PVNGS transitioned to the improved Combustion Engineering (CE) Standard Technical Specifications (NUREG-1432, Revision 1), the LCO was changed to require both the gaseous and particulate channels be operable in LCO 3.4.16. The containment sump instrumentation LCO requirements remained unchanged. It was the APS position, that while the gaseous channel was less timely in the detection of RCS leakage, due to improved fuel performance, it remained an important element of the system design to comply with the intent of Regulatory Guide (RG) 1.45, and the past NRC safety evaluations regarding General Design Criteria (GDC) 4 and approval of leak-before-break methodology for PVNGS.

Subsequent licensees have requested, and the NRC staff approved, the removal of the gaseous radiation instrumentation from the RCS leak detection instrumentation LCO requirements (Reference 1). In addition, TSTF-513 was approved and adopted by various licensees (e.g., D. C. Cook, Comanche Peak, ANO, Farley, etc.), to address the reduced timeliness of the gaseous instrumentation in detecting RCS leakages as a result of improved fuel performance.

2.5 RCS Leak Detection System Design and Operation

The design and operation of the RCS leak detection system is provided in UFSAR Section 5.2.5, *Reactor Coolant Pressure Boundary Leakage Detection Systems*, and summarized in TS Bases 3.4.16.

GDC 30 of Appendix A to 10 CFR 50 requires means for detecting and, to the extent practical, identifying the location of the source of RCS leakage. Regulatory Guide 1.45 describes acceptable methods for selecting leakage detection systems. Leakage detection systems must have the capability to detect significant Reactor Coolant Pressure Boundary (RCPB) degradation as soon after occurrence as practical to minimize the potential for propagation to a gross failure. Thus, an early indication or warning signal is necessary to permit proper evaluation of all unidentified leakage.

Industry practice has shown that water flow changes of 0.5 gallons per minute (gpm) to 1.0 gpm can readily be detected in contained volumes by monitoring changes in water level, in flow rate, or in the operating frequency of a pump. The containment sump monitor consists of instrumentation used to monitor containment sump level and flow (pump run time). The containment sump used to collect unidentified leakage is instrumented to alarm if the rate of level increase corresponds to a sump inflow greater than 1 gpm for 1 hour. This sensitivity is acceptable for detecting increases in unidentified leakage.

The reactor coolant contains radioactivity that, when released to the containment, can be detected by radiation monitoring instrumentation. Reactor coolant radioactivity levels will be low during initial reactor startup and for a few weeks thereafter until activated corrosion products have been formed and fission products appear from fuel element cladding contamination or cladding defects. Instrument sensitivities of 10^{-9} μCi/cc radioactivity for particulate monitoring and of 10^{-6} μCi/cc radioactivity for gaseous monitoring are practical for these leakage detection systems. Radioactivity detection systems (RU-1) are included for monitoring both particulate and gaseous activities because of their sensitivities and responses to RCS leakage.

RU-1 is used to detect an increase in RCS leakage but not to quantify leakage. As described in UFSAR Sections 5.2.5.3.3, 5.2.5.3.4, and 5.2.5.5, in the event of a high alarm or increasing trend, station procedures direct the operator to perform a water inventory balance within 1 hour to determine the equivalent RCS leak rate.

RU-1 includes a raw count rate channel, which can provide early indication of RCS leakage, as the channel response time is faster than the other channels. However, the raw count rate channel is not required by this LCO.

An increase in humidity of the containment atmosphere would indicate release of water vapor to the containment. Dew point temperature or relative humidity measurements can thus be used to monitor increasing humidity levels of the containment atmosphere as an indicator of potential RCS leakage. Since the humidity level is influenced by several factors, a quantitative evaluation of an indicated leakage rate by this means may be questionable and should be compared to observed increases in liquid flow into or from the containment sump. Humidity level monitoring is considered most useful as an indirect alarm or indication to alert the operator to a potential problem. Humidity monitors are not required by this LCO.

Air temperature and pressure monitoring methods may also be used to infer unidentified leakage to the containment. Containment temperature and pressure fluctuate slightly during plant operation, but a rise above the normally indicated range of values may indicate RCS leakage into the containment. The relevance of temperature and pressure measurements are affected by containment free volume and, for temperature, detector location. Alarm signals from these instruments can be valuable in recognizing a sizable leakage to the containment. Temperature and pressure monitors are not required by this LCO.

The LCO is satisfied when monitors of diverse measurement means are available. Thus, the containment sump monitor in combination with a particulate and gaseous radioactivity monitor (RU-1) provides an acceptable minimum. Either the audible or visual alarm at the control room radiation monitoring system station can be used to satisfy the requirements for alarm functionality of RU-1.

The NRC staff reviewed the PVNGS RCS leak detection system design and concluded in NUREG-0857, Section 5.2.5, *Reactor Coolant Pressure Boundary Leakage Detection*:

> "Based on the above, the staff concludes that the RCPB (reactor coolant pressure boundary) leakage detection systems are diverse and provide reasonable assurance that primary system leakage (both identified and unidentified) will be detected. The systems meet the requirements of GDC 30 with respect to provisions for RCPB leak detection and identification, and the guidelines of RG 1.45 with respect to RCPB leakage detection system design and are, therefore, acceptable. The staff further concludes that the [Combustion Engineering Standard Safety Analysis Report] CESSAR interface requirements, as discussed in CESSAR [Safety Evaluation Report] SER Section 5.2.5, are satisfied by the above-described design."

2.6 Technical Specification and Procedures Interface

In accordance with Surveillance Procedures 40ST-9ZZM1, 40ST-9ZZM2, 40ST-9ZZM3, and 40ST-9ZZM4 (Section 6.1.3 and Appendix B), operations performs containment sump inventory and discharge balance, also called containment sump in-leakage, every shift (once per 12 hours) while in Modes 1, 2, 3, and 4. If there is evidence of significant leakage into the Reactor Cavity Sump (note that this does not necessarily mean TS 3.4.14 actions statements have been entered, as the leakage could still meet TS requirements), a calculation of RCS water inventory is performed in accordance with 40ST-9RC02 (preferred), 40ST-9RC05 (manual), or 40ST-9RC08 (backup). This is the same calculation of RCS water inventory [Surveillance Requirement (SR) 3.4.14.1] that is required per LCO 3.4.16 when the required containment sump monitor is inoperable (performed once per 24 hours) or the required containment atmosphere radioactivity monitor is inoperable and performance of SR 3.4.14.1 is selected instead of analyzing grab samples from the containment atmosphere (performed once per 24 hours).

While at a steady state in Modes 1, 2, 3, and 4, calculation of RCS water inventory is required to be performed every 72 hours and has a minimum performance time of approximately every 6 hours given the 4 hour wait after dilution of boration. As such, one of the monitors being inoperable per LCO 3.4.16 potentially increases a surveillance that was already being performed from every 72 hours (at a minimum) to every 24 hours. In accordance with the proposed TS change, both monitors being inoperable reduces the surveillance or grab sample from 24 hours to 12 hours, in addition to requiring a monitor be restored within 72 hours versus the current 30 days with one monitor inoperable or LCO 3.0.3 entry for both monitors out of service. Moreover, if significant sump leakage had already been detected via other surveillances (such as the containment sump inventory and discharge balance performed every shift), the containment sump rate increases would have already been evaluated.

It is lastly noted that 40ST-9RD01 is performed once per hour in Modes 1, 2, 3, and 4 when the containment sump level and flow monitoring system is inoperable and the leakage alarm is actuated for the containment sumps, until the total sump in-leakage is reduced to less than 1 gpm. Although performance of this procedure only implements the monitoring requirements of UFSAR 5.2.5.5, *Evaluation of Reactor Coolant Leakage Indication*, and is

not a substitute for operable RCS leak detection instrumentation as described in LCO 3.4.16, it provides further confirmation of the timely response and monitoring of leakage that occurs even when the required sump monitor is inoperable.

2.7 Leak-Before-Break Methodology Discussion

Leak-before-break (LBB) is a concept that certain piping material has sufficient fracture toughness (i.e., ductility) to resist rapid flaw propagation. A postulated flaw in such piping would not lead to pipe rupture and potential damage to adjacent safety related systems, structures and components before the plant could be placed in a safe, shutdown condition. Before pipe rupture, the postulated flaw would lead to limited but detectable leakage, which would be identified by the leak detection systems in time for the operator to take action.

Regarding application of the GDC 4 LLB methodology and its adoption for CE-System 80 plants and PVNGS, in particular, the following information is provided. 10 CFR 50, Appendix A, *General Design Criteria for Nuclear Power Plants*, was initially issued February 20, 1971 (36 FR 3256). General Design Criterion 4 was part of the initial issue of the GDCs.

GDC 4 was later revised twice (April 11, 1986, 51 FR 12501 and October 27, 1987, 52 FR 41288) to reflect LBB methodologies to permit the removal of pipe whip restraints. The removal of pipe whip restraints makes inservice inspections easier and reduces personnel exposure associated with pipe whip restraint movement for examinations of the piping, and thus, enhances safety. PVNGS adopted the revised GDC 4 provisions in the PVNGS Final Safety Analysis Report (FSAR). The NRC recognized and accepted the adoption of LBB in Supplement 11 of the PVNGS NRC SER (NUREG-0857).

Specifically, Supplement 11 of the PVNGS NRC SER (Reference 2) states:

> "3.6.2 Determination of Break Locations and Dynamic Effects Associated With the Postulated Rupture of Piping"
>
> "The applicant has proposed changes in FSAR Section 3.6.2 which would eliminate break locations and dynamic effects associated with the postulated ruptures from the design basis of the reactor coolant system main loop piping. The proposed changes are based on the limited-scope rule on General Design Criterion (GDC) 4 (Appendix A to 10 CFR 50) (51 FR 12502), which permits the exclusion of postulated pipe ruptures and associated dynamic effects in the design of the reactor coolant loop (RCL) piping, provided the conditions for the application of the leak-before-break methodology are satisfied. The applicant has referred to the submittal by Combustion Engineering (CE) under the Combustion Engineering Standard Safety Analysis Report (CESSAR) docket for demonstration that the leak-before-break requirements have been satisfied for CE plants, which includes Palo Verde Units 1, 2, and 3. The staff has previously reviewed and approved the CE analyses which were performed for satisfaction of these requirements. The staff finds this acceptable as the basis for the proposed changes to FSAR Section 3.6.2 for Palo Verde Units 1, 2, and 3.
>
> The staff finds that the proposed changes in the Palo Verde Units 1, 2, and 3 FSAR, Section 3.6.2, reflect the limited-scope rule on GDC-4, and demonstrate that Palo Verde Units 1, 2, and 3 are in compliance with this rule."

Part of this methodology was an expectation that the reactor coolant system pressure boundary leak detection system would be consistent with the guidelines of RG 1.45 so that it can detect leakage of 1 gpm in one hour. In addition, this methodology requires that the calculated leak rate through the postulated flaw is large relative to the required sensitivity of the plant leak detection systems. The margin is at least a factor of 10 for leakage.

Additional margins are applied for leakage-size crack vs. the critical-size crack and for safe shutdown earthquake loads versus limit loads (factor of 3 for each).

As stated previously, the PVNGS FSAR Section 3.6.2 and the NRC SER for PVNGS document the adoption of the updated GDC 4 provisions, which included crediting the RCS leak detection system and it being sufficiently sensitive to detect RCS leakage.

The PVNGS Aging Management Program helps ensure that the analytical bases of the LBB fatigue crack propagation analysis are maintained through cycle count action limits and corrective actions. In short, PVNGS Aging Management Program helps prevent such leaks as described in UFSAR Chapter 19.

2.8 Primary Water Stress Corrosion Cracking (PWSCC) Considerations

After the rulemaking for the revision to GDC 4 was completed and the Standard Review Plan (NUREG-0800) was revised to reflect the LBB methodologies (Section 3.6.3, *Leak-Before-Break Evaluation Procedures*, August 28, 1987) there was industry operating experience (V. C. Summer) that identified PWSCC for Inconel Alloy 82/182 weld metals in the RCS that had not been factored into the existing LBB methodologies. The industry and the NRC established structural weld overlays and other techniques to address this emergent issue (MRP-139, *Materials Reliability Program: Primary System Piping Butt Weld Inspection and Evaluation Guideline,* September 2005).

The Standard Review Plan was revised (Section 3.6.3, Revision 1, March 2007) to allow PWSCC to be considered as an active degradation mechanism in Alloy 600/82/182 welds in pressurized water reactor plants. GDC 4 is relevant in the sense that the RCS leak detection system is credited in supporting the LBB methodologies.

NRC Regulatory Issue Summary (RIS) 2010-07, *Regulatory Requirements for Application of Weld Overlays and Other Mitigation Techniques in Piping Systems Approved for Leak-Before-Break*, dated June 8, 2010, reminded licensees that performed weld overlays that the potential existed to invalidate the NRC approval of LBB for their sites.

APS reviewed the RIS and concluded:

> "At Palo Verde weld overlay on DMW [dissimilar metal welds] have been used on six (6) pressurizer nozzles and three (3) hot leg branch connections. These locations are the RCS nozzle welds for the following lines: Hot Leg to both Shutdown Cooling lines, Hot Leg to Surge line…, Pressurizer Spray and four Pressurizer Relief lines. Leak Before Break (LBB) has not been credited for any of these welds per UFSAR Figures 3.6-11 through 3.6-19. All these locations are not classified as LBB locations; therefore, this RIS does not apply."

As stated above, the original LBB evaluations that were submitted considered stress corrosion cracking mechanisms to be unlikely due to material selection and chemistry control. It is now known, however, that PWSCC is a possible mechanism for nickel-based alloys and may be the mechanism that causes the initial flaw to go through-wall.

In accordance with the Millstone Safety Evaluation [Reference 5], a flaw growth on the order of 0.25 inches is considered significant (this is assumed to be a reasonable acceptance criterion because it is compared to the leakage flaw size of 5 inches). For a flaw to progress to the point of challenging flaw stability, its size must first progress from a short length that causes an incipient detectable leak, to a size that causes 10 gpm leakage. The size that would allow a 10 gpm leak rate (applying the safety factor of 10 for leakage) is on the order

of about 5 inches long for a branch pipe and larger for the loop piping. Even at the 10 gpm size, the LBB methodology ensures a margin of at least 2.0 on the limiting flaw size for stability. Therefore, a small flaw extension that is a small fraction of the 5 inch long, 10 gpm leak size (approximately 0.25 inches), is a reasonable measure of flaw growth significance. Using this measure of acceptable flaw growth, an acceptable duration can be determined once the flaw growth velocity is estimated.

Addressing PWSCC growth, its flaw growth rate (at non-overlayed nickel-based alloy locations) is very slow. Electric Power Research Institute (EPRI) report MRP-55, Figure 5-2, plots a large number of measured crack growth rates against the applied stress intensity factor. All data are bounded by a rate of 1.0 E-09 meters/second, or 1.25 inches/year, even at very high stress intensity factors. At this rate, more than 70 days would be required for the flaw to extend by 0.25 inches. However, the EPRI MRP-115 report provides the flaw growth data for the Alloy 82 and Alloy 182 material. The crack growth rates for Alloy 82 and Alloy 182 as shown in Chapters 4 and 5 of MRP-115 are higher than the growth rate of Alloy 600 material. As such, the flaw growth rate in MPR-115 is used for this assessment. The maximum flaw growth rate of Alloy 182, as shown in Chapter 4 of MRP-115, could be as high as 2E-09 meters/second (8E-08 inches/second or 2.52 inches/year). At this rate, more than 35 days would be required for the flaw to extend by 0.25 inches.

The issue of PWSCC was addressed as part of the PVNGS renewed operating licensing process. The Internals Aging Management Program manages the aging effects (per implementation of EPRI reports MPR-227-A and MPR-228) on the reactor vessel internal components, including various forms of cracking such as PWSCC. The Nickel Alloy aging management program manages cracking due to PWSCC in most plant locations that contain Alloy 600. The Alloy 600 aging management program uses inspections, mitigation techniques, repair/replace activities and monitoring of operating experience to manage the aging of Alloy 600 at PVNGS. Mitigation techniques are implemented when appropriate to preemptively remove conditions that contribute to PWSCC (UFSAR 19.1). These programs have been reviewed by the NRC for the period of extended operation (PEO).

For the PEO, an environmentally assisted fatigue (EAF) factor has been incorporated into the design of Class 1 piping systems. This is to more accurately account for the heat, potential humidity, and radiation found in containment and the effects on the fatigue rate of the piping materials that were not considered in the original design. Even with the added EAF factor, many fatigue cycles would be necessary to grow the postulated flaw from its incipient stage to a length that would challenge flaw stability. Startup/shutdown cycles are infrequent. Most transients affecting a branch line occur in conjunction with a plant shutdown, after which the leakage would be discovered by the normal reactor building walkdown. If an earthquake occurred during normal power operations, it would still contribute only about 50 cycles, not enough for significant flaw growth for a flaw that is less than half the critical flaw size for design basis seismic loads.

2.9 Proposed Completion Times

The proposed required actions for loss of TS credited RCS leak detection instrumentation do not change the LCO requirements or the system design or normal operational practices. The proposal provides an alternative to rapid plant shutdown, for a limited period of time (72 hours) should the RCS leak detection systems credited in the LCO become inoperable, provided alternative RCS leak detection actions are implemented. The proposed completion times (12 hours) for the alternative RCS leak detection required actions are consistent with providing high confidence that an emergent RCS leak will be detected in advance of significant pipe crack propagation.

Specifically:

(1) The 72 hours is appropriate since additional actions will be taken during this limited time period to ensure RCS leakage in excess of the unidentified leakage TS limit of one gpm, will be readily detectable. This provides reasonable assurance that any significant RCS pressure boundary degradation will be detected soon after leak occurrence and therefore minimize the potential for subsequent growth propagation to a gross failure. This is consistent with the requirements of GDC 30 and also Criterion 1 of 10 CFR 50.36(c)(2)(ii), which requires installed instrumentation to detect and indicate in the control room a significant abnormal degradation of the RCBP.

The RCS water inventory balance calculation determines the magnitude of RCS unidentified leakage by use of instrumentation readily available to the control room operators. However, the proposed additional actions will not restore the continuous monitoring capability normally provided by the inoperable equipment. The RCS water inventory balance is capable of identifying a one gpm RCS leak rate. The containment grab samples will also identify an increase in RCS leak rate, which would then be quantified by the RCS water inventory balance. Since these additional actions are sufficient to ensure RCS leakage is within TS limits, it is appropriate to provide a limited time period to restore at least one of the TS-required leakage monitoring systems.

Lastly, operation during this limited time period with both the monitors inoperable will not adversely impact the ability to detect a small RCS leak in sufficient time to prevent excessive flaw propagation consistent with LBB analysis. For PWSCC, the above evaluation has shown that it will take more than 35 days for a flaw or crack to extend its initial length by 0.25 inches. Similarly, for fatigue crack growth, many fatigue cycles would be necessary to grow the flaw from its incipient stage to a length that would challenge flaw stability. Hence, the proposed 72 hour completion time to restore at least one TS leakage detection instrument to service is well within this time frame such that significant crack or flaw propagation is not expected to occur.

(2) Similarly for 12 hours, only negligible flaw growth is possible by the PWSCC mechanism. Also, significant flaw growth due to fatigue in a 12-hour period is credible, but the flaw growth is insignificant because the fatigue crack growth rate is less than the crack growth rate due to PWSCC and that loading due to transients and operational cycles during the 12-hour period will not lead to significant fatigue flaw growth (see discussion in the previous subsection). Therefore, considering fatigue and PWSCC mechanisms, neither is capable of causing a significant flaw extension with the 12-hour monitoring periodicity.

(3) The proposed alternative actions are the same as the current TS Required Actions B.1.1 and B.1.2 and, therefore, are consistent with the PVNGS licensing basis and are similar to the precedent established in Reference 5. The option of containment grab samples is retained to address the potential of the RCS not being in steady state operations. The current PVNGS TS acknowledges the potential for the RCS not being in steady state operations by a Note for SR 3.4.14.1 which permits 12 hours for stabilization of RCS operation before performance of the SR.

3. TECHNICAL EVALUATION

The proposed required actions for loss of TS credited RCS leak detection instrumentation do not change the LCO requirements or the system design, or normal operational practices. The proposal provides an alternative to rapid plant shutdown, for a limited period of time (72 hours) should the RCS leak detection systems credited in the LCO become inoperable,

provided alternative RCS leak detection actions are implemented. The proposed completion times (12 hours) for the alternative RCS leak detection required actions are consistent with providing high confidence that an emergent RCS leak will be detected in advance of significant pipe crack propagation.

Qualitative insights derived from the Palo Verde Probabilistic Risk Assessment (PRA) indicate LOCA events represent a small contribution to Core Damage and Large Early Release frequencies. Assuming the LOCAs modeled in the PRA can adequately bound the contribution of undetected pipe breaks with leakage in excess of the one gpm detection limits, it is concluded that extending the TS 3.4.16 Completion Time to 72 hours has a very low safety significant impact on plant risk.

The NRC staff has indicated that the RCS leak detection issue is low risk (Reference 4).

> "For most component and piping degradation mechanisms, the probability of crack growth resulting in a loss-of-coolant accident (LOCA) prior to detection by one of the reactor coolant system leakage detection systems is extremely low. This conclusion may be drawn due to: (1) the overall diversity and redundancy of the leakage detection systems, (2) relatively slow rates of stable crack growth for most cracking mechanisms, and (3) the low probability of an abnormal loading event (e.g., an earthquake) which could lead to catastrophic failure of the component or piping. Nuclear power plants are designed to provide adequate core cooling following any postulated LOCA up to and including a break equivalent in size to the double ended rupture of the largest pipe in the RCS. This design feature, coupled with the extremely low likelihood of undetected crack growth resulting in a LOCA, leads us to conclude that the risk significance of this issue is low."

Again, the low-risk nature of the loss of the primary RCS leak detection instrumentation, while retaining alternate RCS leak detection actions for a limited time, supports not requiring a rapid plant shutdown, as currently required by TS LCO 3.4.16.

4. REGULATORY EVALUATION

4.1 Applicable Regulatory Requirements

The proposed changes have been evaluated to determine whether applicable regulations and requirements continue to be met.

GDC 30 of Appendix A to 10 CFR 50 requires means for detecting and, to the extent practical, identifying the location of the source of RCS leakage. Regulatory Guide 1.45 describes acceptable methods for selecting leakage detection systems. Leakage detection systems must have the capability to detect significant RCPB degradation as soon after occurrence as practical to minimize the potential for propagation to a gross failure. Thus, an early indication or warning signal is necessary to permit proper evaluation of all unidentified leakage.

As stated in the previous section of this Enclosure, the NRC staff reviewed the PVNGS RCS leak detection system design and concluded in NUREG-0857, Section 5.2.5, *Reactor Coolant Pressure Boundary Leakage Detection*:

> "Based on the above, the staff concludes that the RCPB (reactor coolant pressure boundary) leakage detection systems are diverse and provide reasonable assurance that primary system leakage (both identified and unidentified) will be detected. The systems meet the requirements of GDC 30 with respect to provisions for RCPB leak detection and identification, and the guidelines of RG 1.45 with respect to RCPB

leakage detection system design and are, therefore, acceptable. The staff further concludes that the CESSAR interface requirements, as discussed in CESSAR SER Section 5.2.5, are satisfied by the above-described design."

The proposed license amendment does not propose to change the current PVNGS LCO 3.4.16 or Applicability requirements. The LAR proposes revisions to TS Section 3.4.16, RCS Leakage Detection Instrumentation, required actions, such that a limited time period is provided to restore RCS leak detection instrumentation, in lieu of plant shutdown, so long as alternative RCS leak detection actions are implemented. The proposed completion times (12 hours and 72 hours before plant shutdown) for the alternative RCS leak detection required actions are consistent with providing high confidence that an emergent RCS leak will be detected in advance of significant pipe crack propagation. The technical basis for the proposed completion times is provided in Section 3, *Technical Evaluation*, of this LAR.

4.2 Precedent

The NRC staff reviewed and approved a similar request for the Millstone Power Station, Units 2 and 3 (TAC Nos. MD6640 and MD6641), dated September 30, 2008 [Reference 5]. Millstone requested the removal of the gaseous radiation monitoring function from the RCS leak detection system, which this LAR does not. However, the Millstone application also addressed providing 72 hours for the condition of the loss of all TS credited RCS leakage detection instrumentation, which is consistent with this APS request.

4.3 No Significant Hazards Consideration Determination

Arizona Public Service Company (APS) has evaluated whether a significant hazards consideration is involved with the proposed amendment by focusing on the three standards set forth in 10 CFR 50.92:

1. Does the proposed change involve a significant increase in the probability or consequences of an accident previously evaluated?

Response: No

The proposed change does not make any hardware changes and does not alter the configuration of any plant system, structure, or component (SSC). Therefore, the consequences of an accident are not increased. The RCS leak detection system is not credited for use in the initiation of any automatic protective functions. The TS will continue to require diverse means of leakage detection equipment, thus ensuring that leakage due to cracks would continue to be identified prior to breakage and the plant shutdown accordingly. Additionally, the function of this equipment is not modeled in the probabilistic risk assessment and, therefore, the proposed temporary reliance upon alternative RCS leak detection actions for loss of the credited RCS leak detection instrumentation has no impact on core damage frequency or large early release frequency. Therefore, the probability of occurrence of an accident is not increased.

The proposed change has been evaluated and determined to not increase the probability or consequences of an accident previously evaluated.

2. Does the proposed change create the possibility of a new or different kind of accident from any accident previously evaluated?

Response: No

The proposed change does not involve the use or installation of new equipment, and the currently installed equipment will not be operated in a new or different manner. No new or different system interactions are created, and no new processes are introduced. The

proposed changes will not introduce any new failure mechanisms, malfunctions, or accident initiators not already considered in the design and licensing bases. The proposed change does not affect any SSC associated with an accident initiator.

Based on this evaluation, the proposed change does not create the possibility of a new or different kind of accident from any accident previously evaluated.

3. Does the proposed change involve a significant reduction in a margin of safety?

Response: No

The proposed change does not make any alteration to any RCS leakage detection components. The proposed amendment continues to require diverse means of leakage detection equipment with capability to promptly detect RCS leakage. The proposed temporary reliance upon alternative RCS leak detection actions for loss of the credited RCS leak detection instrumentation provides high confidence that an emergent RCS leak will be detected in advance of significant pipe crack propagation. Additional diverse means of leakage detection capability remain available.

Based on this evaluation, the proposed change does not involve a significant reduction in a margin of safety.

Conclusion

APS concludes that operation of the facility in accordance with the proposed amendment does not involve a significant hazards consideration under the standards set forth in 10 CFR 50.92(c), and, accordingly, a finding of "no significant hazards consideration" is justified. Based on the considerations discussed above, (1) there is reasonable assurance that the health and safety of the public will not be endangered by operation in the proposed manner, (2) such activities will be conducted in compliance with the Commission's regulations, and (3) the issuance of the amendment will not be inimical to the common defense and security or the health and safety of the public.

5. ENVIRONMENTAL EVALUATION

APS has determined that the proposed amendment would change requirements with respect to the use of a facility component located within the restricted area, as defined by 10 CFR 20, or would change an inspection or SR. However, the proposed amendment does not involve (i) a significant hazards consideration, (ii) a significant change in the types or significant increase in the amounts of any effluent that may be released offsite, or (iii) a significant increase in individual or cumulative occupation radiation exposure. Accordingly, the proposed amendment meets the eligibility criterion for a categorical exclusion set forth in 10 CFR 51.22(c)(9). Therefore, pursuant to 10 CFR 51.22(b), no environmental impact statement or environmental assessment need be prepared in connection with the proposed amendment.

6. REFERENCES

1. With respect to the removal of the containment atmosphere gaseous radioactivity monitors, the NRC approved license amendments for South Texas Project, Units 1 and 2, (TAC Nos. MC7258 and MC7259), on October 17, 2005, for Byron Station, Units 1 and 2, and Braidwood Station, Units 1 and 2, (TAC Nos. MC0509, MC0510, MC0507, and MC0508), on January 14, 2005 and for Millstone Power Station, Units 2 and 3 (TAC Nos. MD6640 and MD6641), dated September 30, 2008

2. NRC Safety Evaluation Report for PVNGS (NUREG-0857), Section 5.2.5, *Reactor Coolant Pressure Boundary Leakage Detection,* and Supplement 11 for Section 3.6.2, *Determination of Break Locations and Dynamic Effects Associated With the Postulated Rupture of Piping*
3. CESSAR SER (NUREG-0852) and Supplements
4. NRC Letter ADAMS Accession Numbers ML021750003 and ML021750004, dated July 18, 2002
5. Millstone Power Station, Unit Nos. 2 and 3 – Issuance of Amendment RE: Technical Specifications Regarding Reactor Coolant System Leakage Detection Systems (TAC Nos. MD6640 and MD6641), ML082261529, dated September 30, 2008

ATTACHMENT 1

Technical Specification Pages Mark-ups

Affected Pages:

3.4.16-2
3.4.16-3

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
B. Required containment atmosphere radioactivity monitor inoperable.	B.1.1 Analyze grab samples of the containment atmosphere. OR B.1.2 Perform SR 3.4.14.1. AND B.2 Restore required containment atmosphere radioactivity monitor to OPERABLE status.	Once per 24 hours Once per 24 hours 30 days
~~D~~C. All required monitors inoperable.	~~D.1 Enter LCO 3.0.3~~ C.1 Analyze grab samples of the containment atmosphere. OR C.2 Perform SR 3.4.14.1. AND C.3 Restore at least one RCS leakage detection monitor to OPERABLE status.	~~Immediately~~ Once per 12 hours Once per 12 hours 72 hours
~~C~~D. Required Action and associated Completion Time not met.	~~C~~D.1 Be in MODE 3. AND ~~C~~D.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.16.1	Perform CHANNEL CHECK of the required containment atmosphere radioactivity monitor.	In accordance with the Surveillance Frequency Control Program
SR 3.4.16.2	Perform CHANNEL FUNCTIONAL TEST of the required containment atmosphere radioactivity monitor.	In accordance with the Surveillance Frequency Control Program
SR 3.4.16.3	Perform CHANNEL CALIBRATION of the required containment sump monitor.	In accordance with the Surveillance Frequency Control Program
SR 3.4.16.4	Perform CHANNEL CALIBRATION of the required containment atmosphere radioactivity monitor.	In accordance with the Surveillance Frequency Control Program

ATTACHMENT 2

Clean Technical Specification Pages

Affected Pages:

3.4.16-2
3.4.16-3

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
B. Required containment atmosphere radioactivity monitor inoperable.	B.1.1 Analyze grab samples of the containment atmosphere. OR B.1.2 Perform SR 3.4.14.1. AND B.2 Restore required containment atmosphere radioactivity monitor to OPERABLE status.	Once per 24 hours Once per 24 hours 30 days
C. All required monitors inoperable.	C.1 Analyze grab samples of the containment atmosphere. OR C.2 Perform SR 3.4.14.1. AND C.3 Restore at least one RCS leakage detection monitor to OPERABLE status.	Once per 12 hours Once per 12 hours 72 hours
D. Required Action and associated Completion Time not met.	D.1 Be in MODE 3. AND D.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.16.1	Perform CHANNEL CHECK of the required containment atmosphere radioactivity monitor.	In accordance with the Surveillance Frequency Control Program
SR 3.4.16.2	Perform CHANNEL FUNCTIONAL TEST of the required containment atmosphere radioactivity monitor.	In accordance with the Surveillance Frequency Control Program
SR 3.4.16.3	Perform CHANNEL CALIBRATION of the required containment sump monitor.	In accordance with the Surveillance Frequency Control Program
SR 3.4.16.4	Perform CHANNEL CALIBRATION of the required containment atmosphere radioactivity monitor.	In accordance with the Surveillance Frequency Control Program

ATTACHMENT 3

Technical Specification Bases Page Mark-up
(provided for information only)

Affected Page:

B 3.4.16-4

BASES

ACTIONS (continued)

B.1.1, B.1.2, and B.2

With either the gaseous or particulate containment atmosphere radioactivity monitoring instrumentation channels inoperable, alternative action is required. Either grab samples of the containment atmosphere must be taken and analyzed, or water inventory balances, in accordance with SR 3.4.14.1, must be performed to provide alternate periodic information. With a sample obtained and analyzed or an inventory balance performed every 24 hours, the reactor may be operated for up to 30 days to allow restoration of both of the radioactivity monitors.

The 24 hour interval provides periodic information that is adequate to detect leakage. The 30 day Completion Time recognizes at least one other form of leakage detection is available.

C.1, C.2, and C.3

If all required monitors are inoperable, no automatic means of monitoring leakage are available. Frequent use of indirect methods of monitoring RCS leakage must be implemented. Either grab samples of the containment atmosphere must be taken and analyzed or a water inventory balance must be performed every 12 hours to provide alternate periodic information. With a sample obtained and analyzed or a water inventory balance (SR 3.4.14.1) performed every 12 hours, 72 hours is provided to restore at least one RCS leakage detection monitor. The 72 hour Completion Time is reasonable, considering the low probability of significant RCS leakage occurring during this time and the avoidance of a plant shutdown in response to the loss of monitoring equipment, while providing a reasonable time to restore a monitor to OPERABLE status.

D.1, and D.2

If any Required Action of Condition A or B or C cannot be met within the required Completion Time, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

(continued)